

IAASSE Member Conversion Prediction Model Project

Week 2 Report

Exploratory Data Analysis (EDA), Data Cleaning & Key Feature Identification

Dataset: Telco Customer Churn

Tools: Python, Pandas, Scikit-learn

Reporting Period: September 2026

Executive Summary

This report presents the Week 2 deliverables of the IAASSE Data Science Classification Project. The primary objective of this phase was to systematically explore, clean, and analyze the Telco Customer Churn dataset in preparation for predictive model development in Week 3.

The dataset contains **7,043 customer records and 21 variables**, with customer churn (`Churn`) serving as the target variable. A structured Exploratory Data Analysis (EDA) workflow was conducted to assess data quality, identify patterns, examine potential churn drivers, and determine the most relevant predictive features.

The analysis included initial dataset profiling, data-type validation, missing-value treatment, duplicate detection, categorical preprocessing, univariate analysis, bivariate analysis, correlation analysis, and mutual information-based feature assessment.

A significant data-quality issue was identified in the `TotalCharges` variable, which was incorrectly loaded as an object data type because 11 records contained blank strings. These values were converted to numeric format and subsequently imputed using the median. The `customerID` variable was removed because it functions solely as a unique identifier and does not provide meaningful predictive information.

The analysis identified **Contract, Tenure, OnlineSecurity, TechSupport, and PaymentMethod** as the five highest-impact predictive features. These findings were supported through a combination of correlation analysis and mutual information scoring.

The target variable exhibits moderate class imbalance, with **73.5% of customers retained and 26.5% churned**. Therefore, Week 3 model development should prioritize class-imbalance handling and evaluate models using precision, recall, and F1-score in addition to accuracy.

Overall, the dataset has been successfully cleaned, encoded, analyzed, and prepared for the next stage of the project: predictive model development.

Key Outcomes

- **7,043** customer records analyzed.
- **21** original columns reviewed.
- **11** hidden blank values identified in `TotalCharges`.
- **0** duplicate records detected.
- **0** missing values remaining after preprocessing.
- `customerID` removed as a non-predictive identifier.
- **Top five predictive features:** Contract, Tenure, OnlineSecurity, TechSupport, and PaymentMethod.
- Dataset prepared for Week 3 model training.

Dataset Overview

The Telco Customer Churn dataset consists of **7,043 customer records and 21 variables**. The target variable, `Churn`, indicates whether a customer discontinued the service during the relevant period.

The dataset contains numerical, categorical, demographic, service-related, and account-related variables. Service-related attributes represent the largest feature group, reflecting the importance of service adoption and customer engagement within the telecommunications business model.

Dataset Summary

Category	Details
Total Records	7,043
Original Features	21
Numerical Features	3
Categorical Features	18
Target Variable	Churn
Target Classes	Yes / No
Churn Rate	26.5%
Non-Churn Rate	73.5%

Major Feature Groups

Numerical Variables

- `tenure`
- `MonthlyCharges`
- `TotalCharges`

Demographic Variables

- gender
- SeniorCitizen
- Partner
- Dependents

Service Variables

- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies

Account Variables

- Contract
- PaperlessBilling
- PaymentMethod

Data Cleaning and Preprocessing

A comprehensive data-quality assessment was performed before conducting feature analysis.

Issue Identified in TotalCharges

The TotalCharges column was initially interpreted as an object/string variable rather than a numerical variable. Further investigation showed that **11 records contained blank strings (" ") instead of conventional missing-value indicators.**

This issue could have affected downstream operations such as correlation analysis, scaling, and model training. Therefore, the column was explicitly converted to numeric format.

Cleaning Procedures

Cleaning Procedure	Action	Result
Data-Type Correction	Converted TotalCharges using <code>pd.to_numeric(..., errors='coerce')</code>	Numeric float64 format
Missing-Value Treatment	Median imputation	11 missing values resolved

Cleaning Procedure	Action	Result
Identifier Removal	Removed <code>customerID</code>	Non-predictive identifier eliminated
Categorical Standardization	Converted <code>SeniorCitizen</code> from 0/1 to No/Yes	Consistent representation
Duplicate Detection	Checked for duplicate rows	0 duplicates found

Final Data Quality

Following preprocessing, the dataset contained:

- **No remaining missing values**
- **No duplicate records**
- Correct data types
- **20 model-ready features**
- **All 7,043 original observations retained**

No customer records were removed during the cleaning process.

Target Variable Analysis

The target variable `Churn` shows a moderate class imbalance:

Customer Status	Percentage
No Churn	73.5%
Churn	26.5%

This represents an approximate **2.77:1 imbalance ratio** between non-churned and churned customers.

Although this imbalance is manageable, it creates an important consideration for predictive modeling. A model predicting every customer as "No Churn" could achieve approximately 73.5% accuracy while failing to identify any actual churners.

Consequently, Week 3 model evaluation should place particular emphasis on **precision, recall, and F1-score for the churn class**, rather than relying solely on accuracy.

Potential imbalance-handling strategies include:

- `class_weight='balanced'`
- SMOTE oversampling
- Stratified train-test splitting
- Stratified cross-validation

Key Findings from Exploratory Data Analysis

Contract Type

Contract type emerged as the strongest churn indicator.

Contract Type	Churn Rate	Customer Count
Month-to-month	42.7%	3,875
One year	11.3%	1,473
Two year	2.8%	1,695

Customers on month-to-month contracts demonstrate substantially higher churn than customers on longer-term contracts.

The difference between month-to-month and two-year customers indicates that contract commitment is a major factor associated with customer retention.

Service Subscription

Service adoption also demonstrated strong relationships with churn.

Service	Without Service	With Service	Difference
OnlineSecurity	41.8%	14.6%	27.2 pp
TechSupport	41.6%	15.2%	26.4 pp
OnlineBackup	39.9%	21.5%	18.4 pp
DeviceProtection	39.1%	22.5%	16.6 pp

OnlineSecurity and TechSupport demonstrated the largest observed differences, indicating that these services may represent important retention opportunities.

Tenure

Customer tenure showed a strong negative relationship with churn.

Customers with **less than 12 months of tenure account for nearly half of all churn**, indicating that the first year represents an especially important retention period.

The correlation between tenure and churn was reported as $r = -0.35$, supporting its position as one of the strongest individual predictors in the analysis.

Payment Method

Payment method also demonstrated substantial differences in churn rates.

Payment Method	Churn Rate
Electronic check	45.3%
Mailed check	19.1%
Bank transfer (automatic)	16.7%
Credit card (automatic)	15.2%

Customers using electronic checks showed the highest churn rate at **45.3%**, considerably higher than customers using automatic payment methods.

Demographic Risk Factor

Senior-citizen status was identified as an additional risk factor. The analysis reports approximately **42% churn among senior customers compared with 24% among non-senior customers**.

This finding suggests that demographic characteristics may interact with other risk factors.

The highest-priority retention segment identified in the analysis consists of customers who combine multiple risk characteristics, particularly:

- Low tenure
- Month-to-month contracts
- No OnlineSecurity
- No TechSupport
- Senior-citizen status

Feature Importance Ranking

Feature importance was assessed using a combination of **absolute correlation** and **mutual information**.

Correlation measures linear association, while mutual information can capture broader non-linear dependencies. Using both measures provides a more comprehensive basis for feature prioritization.

Rank	Feature	Impact	Correlation	Mutual Information
1	Contract	High	0.40	0.098
2	Tenure	High	0.35	0.089
3	OnlineSecurity	High	0.29	0.052
4	TechSupport	High	0.28	0.048
5	PaymentMethod	High	0.11	0.041
6	InternetService	Medium	0.15	0.038
7	MonthlyCharges	Medium	0.19	0.035
8	TotalCharges	Medium	0.20	0.032
9	OnlineBackup	Medium	0.22	0.028
10	PaperlessBilling	Medium	0.19	0.025

The results identify **Contract, Tenure, OnlineSecurity, TechSupport, and PaymentMethod** as the primary features for consideration during Week 3 model development.

Business Recommendations

Based on the EDA findings, the following actions are recommended:

Encourage Long-Term Contracts

Month-to-month customers demonstrate substantially higher churn than customers with two-year contracts.

Recommended action: Introduce graduated incentives for customers who transition to one- or two-year contracts.

Strengthen First-Year Customer Onboarding

The first 12 months represent a particularly high-risk period.

Recommended action: Implement a structured "First 90 Days" customer-success program featuring proactive check-ins, usage guidance, and early-warning indicators.

Promote Security and Technical Support Services

Customers without OnlineSecurity and TechSupport demonstrate substantially higher churn rates.

Recommended action: Provide introductory offers or bundled packages for these services to encourage adoption.

Encourage Automatic Payment Methods

Electronic-check users have the highest observed churn rate among payment methods.

Recommended action: Encourage customers to migrate toward automatic bank transfers or credit-card payments through appropriate incentives.

Provide Targeted Support for Senior Customers

The analysis identifies higher churn among senior customers.

Recommended action: Develop simplified billing communication and accessible technical-support options for this customer segment.

Review Pricing for New Customers

Higher monthly charges are associated with increased churn, particularly among customers with shorter tenure.

Recommended action: Evaluate targeted loyalty incentives for newer customers with relatively high monthly charges.

Challenges and Solutions

Challenge	Root Cause	Solution	Key Lesson
Hidden Missing Values	Blank strings in <code>TotalCharges</code>	Numeric conversion followed by median imputation	Always validate data types after loading
Feature Selection Bias	Correlation captures primarily linear relationships	Combined correlation with mutual information	Use multiple feature-importance techniques
Categorical Encoding	Label encoding can introduce artificial ordinal relationships	Used encoding for exploratory analysis and planned one-hot encoding for modeling	Separate analytical encoding from modeling encoding

Week 2 Deliverables

Deliverable	Status	Output
Full EDA Notebook	Complete	Week2_EDA_Customer_Churn.ipynb
Cleaned Dataset	Complete	cleaned_churn_data.csv
Modeling-Ready Dataset	Complete	encoded_churn_data.csv
Feature Importance Ranking	Complete	Section 5
Business Recommendations	Complete	Section 6

Week 3 Preparation

With data cleaning, exploratory analysis, and feature ranking completed, the project is ready to transition from exploration to predictive modeling.

The Week 3 plan includes:

Model Development

Train and compare:

- Logistic Regression
- Decision Tree
- Random Forest

Class-Imbalance Handling

Evaluate:

- `class_weight='balanced'`
- SMOTE oversampling

Feature Engineering

Investigate derived variables such as:

```
AvgMonthlyCharge = TotalCharges / tenure
```

Train-Test Strategy

Use a **stratified 80/20 train-test split** to maintain the target-class distribution.

Cross-Validation

Apply **5-fold stratified cross-validation** to obtain more robust performance estimates.

Conclusion

The Week 2 phase successfully established a clean and well-understood foundation for the Telco Customer Churn classification project.

The analysis identified important data-quality issues, resolved missing and incorrectly formatted values, removed the non-predictive customer identifier, and confirmed that the final dataset contains no duplicate or unresolved missing records.

EDA revealed several strong churn patterns, particularly around **contract type, customer tenure, service adoption, and payment method**. The combined feature-ranking approach further identified **Contract, Tenure, OnlineSecurity, TechSupport, and PaymentMethod** as the five highest-priority predictive features.

The findings provide both a strong technical foundation for Week 3 model development and actionable insights for customer-retention strategies.

The dataset and analytical pipeline are therefore **ready for predictive model training and evaluation in Week 3**.